MODULE-II

1. Distinguish between CISC vs RISC Design Philosophy.

RISC	CISC
RISC stands for Reduced Instruction set computer	CISC stands for Complex Instruction set Computer
It is a microprocessor architecture that uses small instruction set of uniform length.	This offers hundreds of instructions of different sizes to the users.
These simple instructions are executed in one clock cycle.	This architecture has a set of special purpose circuits which help execute the instructions at a high speed.
These chips are relatively simple to design	These chips are complex to design.
They are inexpensive	They are relatively expensive
Examples of RISC chips include SPARC, POWER PC	Examples of CISC includes Intel Architecture, AMD.
It has less number of instructions.	It has more number of instructions.
It has fixed-length encoding for instructions	It has variable length encoding of instructions.
Simple addressing formats are supported	The instructions interact with memory using complex addressing modes.
It doesn't support arrays.	It has a large number of instructions. It supports arrays.

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It doesn't use condition codes	Condition codes are used.
Registers are used for procedure arguments and return addresses.	The stack is used for procedure arguments, and return addresses.

2. Distinguish between Von-Neumann and Harvard architecture.

Von Neumann Architecture	Harvard Architecture
It is ancient computer architecture based on stored program computer concept.	It is modern computer architecture based on Harvard Mark relay based model.
Same physical memory address is used for instructions and data	Separate physical memory address is used for instructions and data.
There is a common bus for data and intruction transfer.	Separate buses are used for transferring data and instructions.
Two clock cycles are required, to execute single instruction.	An instruction is executed in a single cycle.
It is cheaper in cost	It is costly than Von Neumann Architecture.
CPU can not access instructions and read/write at the same time.	CPU can access instructions and read / write at the same time.
It is used in personal and small computers	It is used in micro controllers & signal processing.

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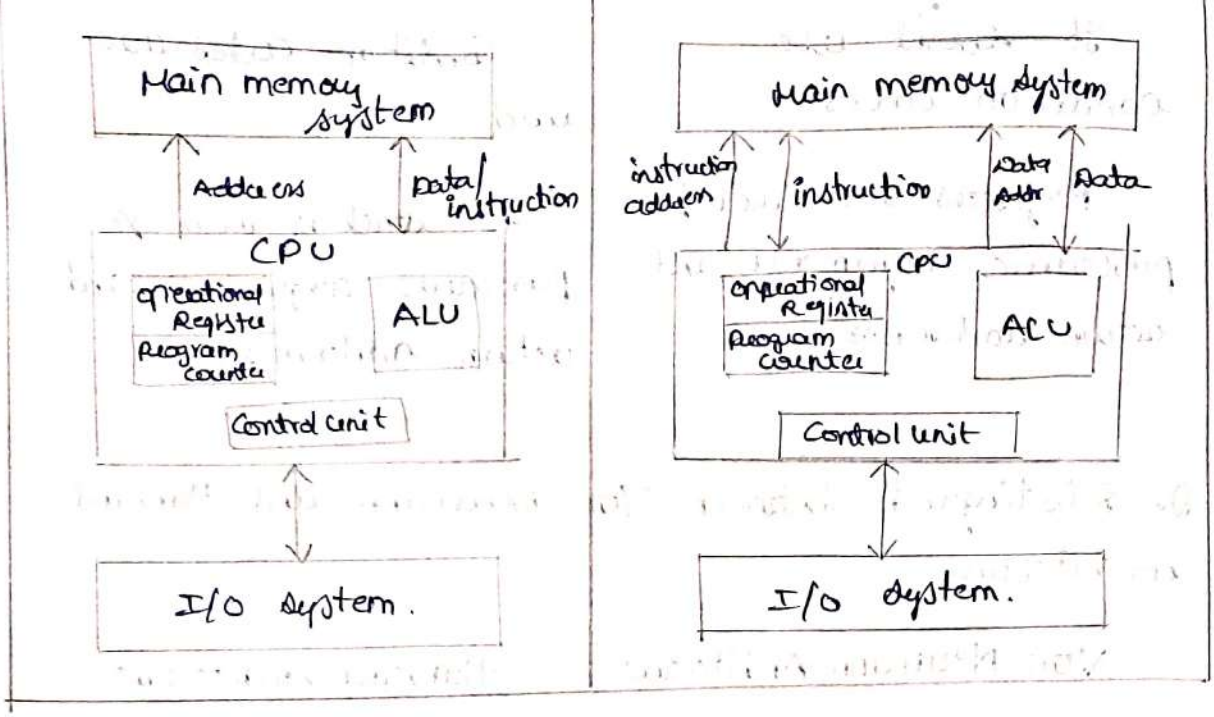
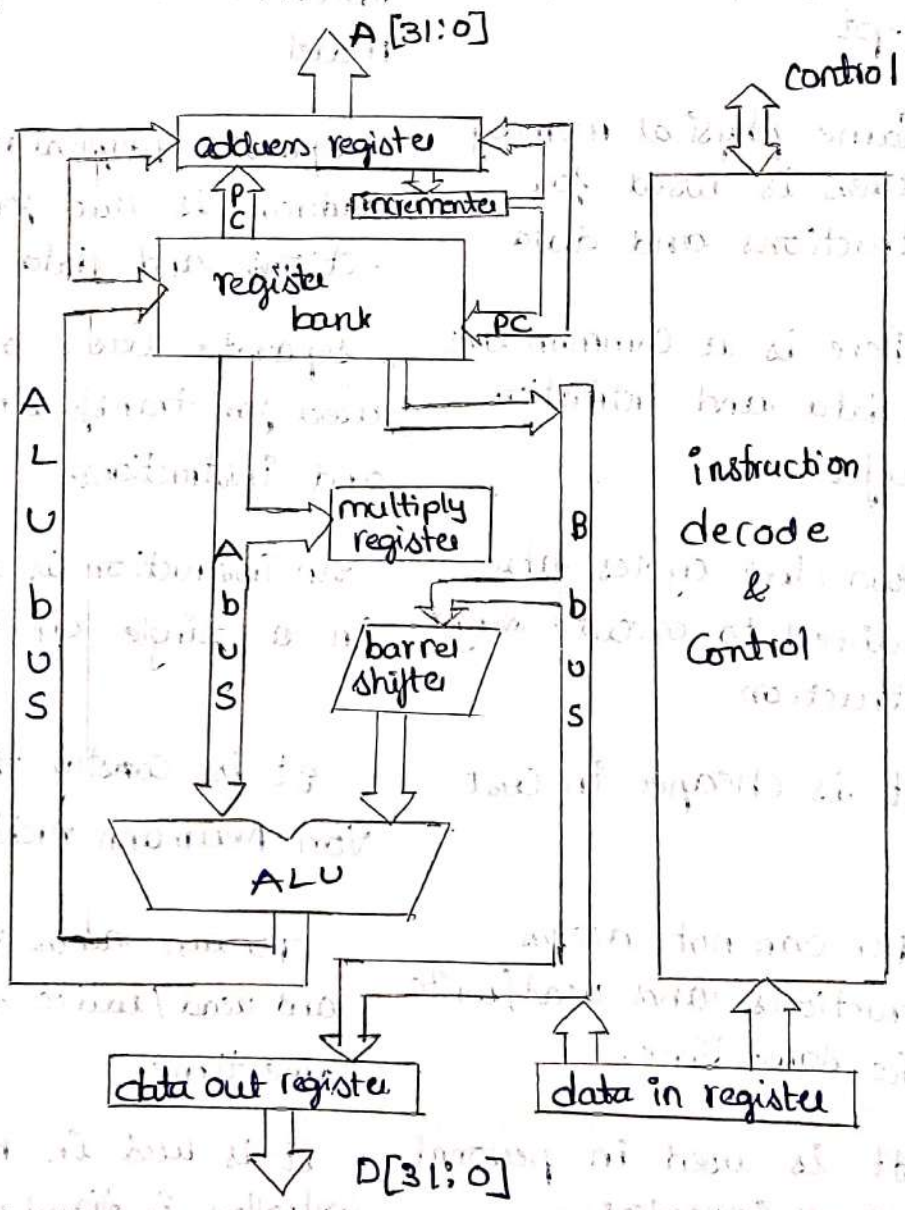


3. Draw and Explain ARM Architecture.

ARM Architecture



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The ARM (Advanced RISC Machine) architecture is based on the reduced instruction set computer (RISC) design philosophy, which focuses on simplicity and efficiency.

From the Architecture diagram we can see various components.

1. Address Register:-
   - Holds memory addresses for data and instructions.
   - Works with the Program counter for instruction fetching.

2. Incrementer:
   Updates the program Counter (PC) to point to the next instruction in memory.

3. Register Bank:
   - Contains general-purpose registers used for data storage during program execution.
   - Registers are central to the architecture, offering fast access to operands.

4. Multiply Register:
   Specialized hardware for handling multiplication operations efficiently.

5. Barrel Shifter:
   - performs shift and rotate operations on data.
   - Enables complex instructions like multiplication by powers of two.

6. ALU (Arithmetic Logic Unit):
   - performs arithmetic (addition, subtraction, etc.) and logical (AND, OR, XOR, etc) operations.
   - Core computational unit of the architecture

7. Data Out Register:
   - Temporarily stores data to be written to memory or output devices.

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8. Data In Register:
Temporarily stores data read from memory or input devices.

9. Instruction Decode and Control Unit:
- Decodes instructions fetched from memory;
- Controls the data flow and manages interactions between components.

10. Buses:
- A Bus and B Bus: Transfer operands from the register bank to other units (e.g ALU, barrel shifter).
- ALU Bus: Transfers results from the ALU to other components.
- D[31:0]: Data bus for transferring 32-bit data between the processor and memory.

Key points:
→ It supports 32-bit data processing
→ Operands must be loaded into registers before processing, and results must be stored back into memory.
→ Designed for fast execution and low power consumption
→ Many instructions can be conditionally executed, reducing branch overhead.
→ Improves performance by overlapping instruction execution (fetch, decode and execute) pipeline Design.

ARM processors are widely used in embedded systems, mobile devices, and low-power applications due to their energy efficiency and performance.

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4. Draw and Explain Cortex-M-Series

The Cortex-M series is a family of 32 bit microcontroller processors optimized for embedded and real-time applications. They are based on ARMv7-M and ARMv8-M architectures, offering excellent performance, low power consumption and scalability.

Cortex-M3:-

1. General purpose CPU:

Designed for microcontroller applications with optimized debug features.

2. Thumb-2 Instruction set

Combines ARM 32-bit and Thumb 16 bit instruction for high code density and efficiency.

Key features:-

→ Supports hardware divide and multiply instructions for performance.

→ Two operating modes:

- Thread Mode for user programs
- Handler Mode privileged, for exception handling.

→ NVIC (Nested Vectored Interrupt Controller):

- Supports up to 240 interrupts with 256 priority levels.
- Enables fast interrupt handling with hardware stacking.

→ 32-bit core with 18 registers: r0-r15, XPSR, and a second stack pointer for handler mode.

→ Unaligned Data Access:

previously limited to high-end processors.

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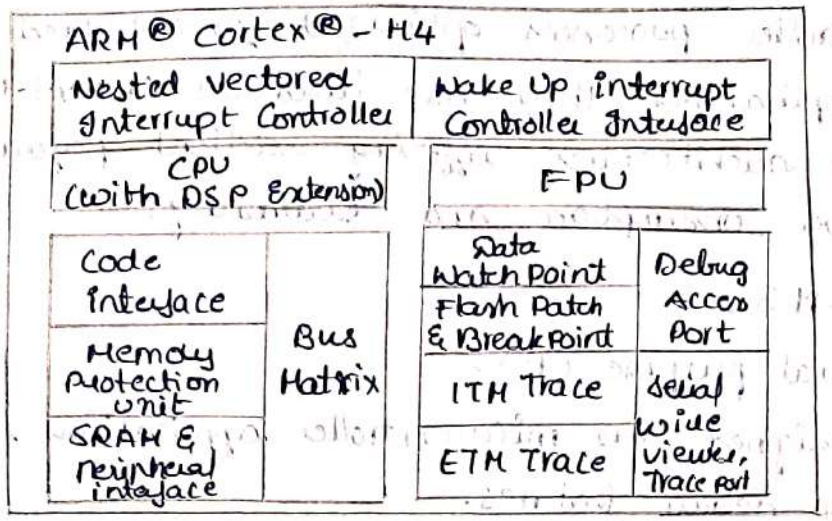
Cortex - M4 :-



1. Enhancements over Cortex - M3 :
   - Includes DSP extensions for signal processing.
   - Implements 72 DSP instructions, including single-cycle multiply-accumulate.

2. Thumb-2 Instruction set :
   Offers ARM's 32-bit performance with Thumb's code density.

3. Floating Point unit (FPU) :
   - IEEE 754-Compliant single precision support for advanced calculations.
   - Useful in signal processing, filtering and compression.

4. Advanced features :
   - Saturating arithmetic : For signal processing applications.
   - Power Management : Sleep and deep sleep modes for energy efficiency.
   - Memory Protection unit : Safeguards memory in safety critical systems.

5. Interrupt Handling :
   Deterministic and low-latency for real-time applications
   Wake-up interrupt controller for low power operations.

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6. Debug capabilities:

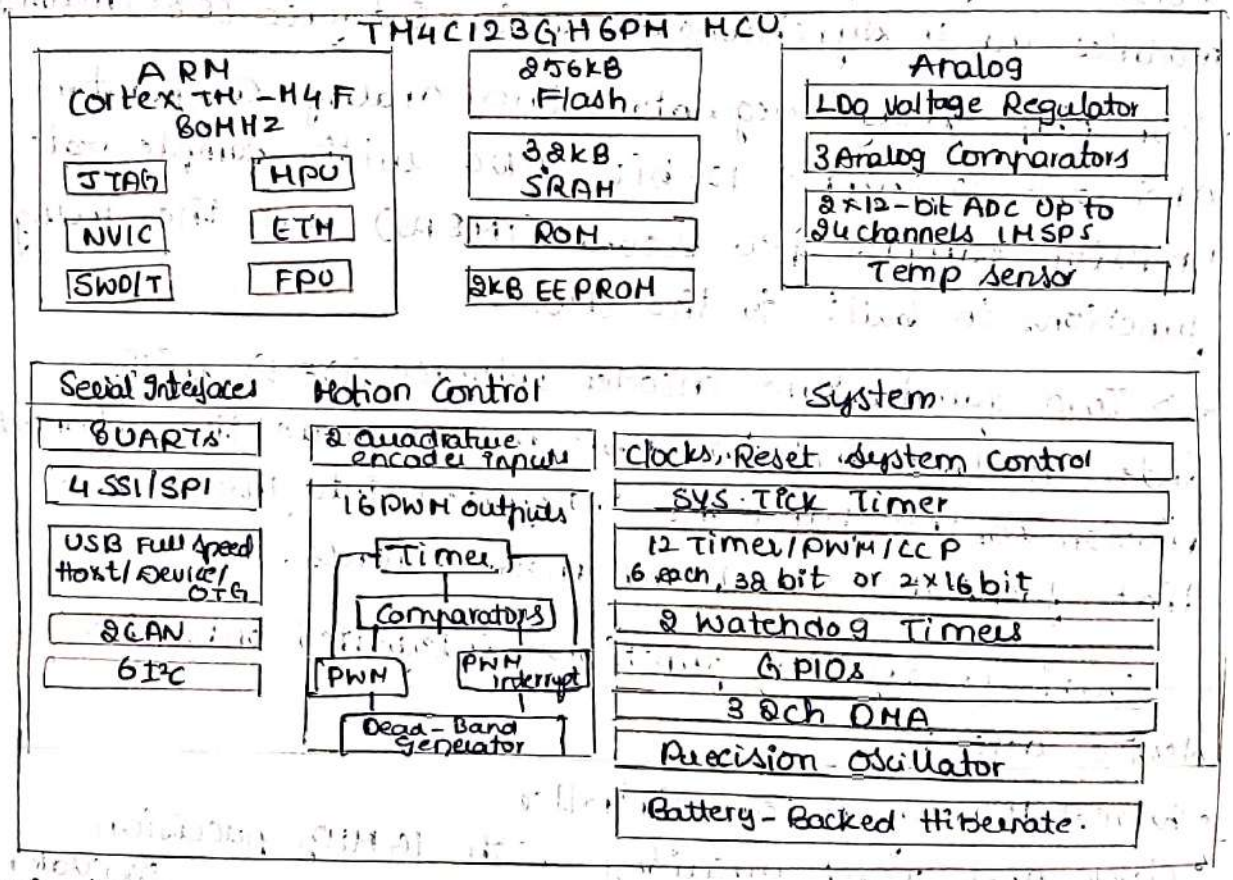
Includes Embedded Trace Macrocell (ETM), Data watch point & Trace (DWT), and Debug Access port (DAP).

Supports hardware breakpoints, code patching, and event tracing.

Applications:-

Ideal for microcontrollers, embedded systems and real-time signal processing due to its energy efficiency and high performance features.

5. Write a short note on TM4C family and its targeted applications.



Features:-

→ TM4C123GH6PM microcontroller has 32 bit ARM Cortex M4 CPU core with 80MHz clock rate.

→ Memory protection unit (MPU) provides protected operating system functionality and floating point unit (FPU) supports IEEE 754 single precision operations.

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→ JTAG / SWD / ETM for serial wire debugs and traces (SWD/T).

→ Nested vector interrupt Controller (NVIC) reduces interrupt response latency.

→ System Control block (SCB) holds the system configuration information.

→ The microcontroller has a set of memory integrated in it: 256KB flash memory, 32 KB SRAM, 2KB EEPROM and ROM loaded with TIVA software library and boot loader.

→ Serial Communications peripherals such as: 2 CAN controllers, full speed USB controller, 8 UARTs, 4 I2C modules and 4 synchronous serial interface (SSI) modules.

→ On chip voltage regulator; two analog comparators and two 12 channel 12-bit ADC with sample rate 1 million samples per second (1MSPS) are the analog functions in built to the device.

→ Two quadrature encoder interface (QEI) with index module and two PWM modules are the advanced motion control functions integrated into the device that facilitate wheel and motor controls.

→ Various system functions integrated into the device are:

- Direct Memory Access Controller
- clock and reset circuitry with 16 MHz precision oscillator
- six 32 bit timers, six 64 bit timers
- 12 32/64 bit captures compare PWM (CCP)
- RTC hibernation module
- 2 Watchdog timers
- 43 GPIOS.

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Few Applications:-

→ Building automation system
→ Lighting Control system
→ Data acquisition system
→ Motion Control
→ IOT and Sensor Networks.

The TM4C123GH6PM microcontroller is targeted for industrial applications, including remote monitoring, electronic point-of-sale machines, test and measurement equipment, network appliances and switches.

6. Write a short note on TM4C129x family and its targeted applications.

TM4C129x			Temperature	85°C	105°C
ARM Cortex-M4 up to 120 MHz			Memory	Power & clocking	
FPU	MPU		up to 1MB Flash	Precision oscillator	
NVIC	ETM	SWD/T	up to 256KB SRAM	RTC Battery-backed hibernation	
			6KB EEPROM	System Modules	
			ROM	8x32-bit Timer/PWM/CCP	
			DMA(32ch)	EPI	
			System Management	LCD	
			1-wire	Systick Timer	
				2x watchdog Timer	
			Debug		
			Real-time JTAG		
Control Peripherals			Comms Peripherals	Analog	
Quadrature Encoder input			8x UART	8x 12ch, 12bit ADCs	
8x PWM Outputs			4x QSSI/SPI	up to 2 MSPS	
			10x I²C	LDO voltage regulator	
Data Protection			2x CAN	3x Analog Comparators	
4x Tamper Inputs			10/100 Ethernet MAC/PHY	Temperature Sensor	
CRC Accelerator			USB Full/High speed (Host/Device/OTG)		
AES, DES, SHA & MD5 Accelerators					

→ TM4C129NCZAD microcontroller has 32 bit ARM Cortex M4F CPU core with 120MHz clock rate.

→ MPU provides a priviledge mode for protected operating system functionality and FPU supports IEEE 754 compliant single precision operations.

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→ JTAG | SWD | ETM for serial wire debug and trace.

→ NVIC reduces interrupt response latency and high performance interrupt handling for time critical application

→ The microcontroller has a set of memory integrated in it: 1MB flash memory, 256KB SRAM, 6KB EEPROM and ROM loaded with TIVA ware software library and boot loader.

→ Serial communications peripherals such as: 2CAN, full speed and high speed USB controller, 8UARTs, 10 I2C modules & 4 synchronous serial interface module.

→ On chip voltage regulator, 3 analog comparators & 2 12 channel 12 bits ADC with 2MSPS and temperature sensor are the analog functions in built to the device.

→ One QEI and 1 PWM module with 8PWM outputs are the advanced motion control functions integrated into the device.

→ Various system functions integrated into the device are: Micro Direct memory access controller, clock and reset circuitry with 16MHz, 8 32-bit timers, low power battery, RTC hibernation module

Applications:-

1. Industrial applications: SCADA, PLC, Motor Control
2. IOT: Smart homes, smart grid, ~~wear~~ wearable devices
3. Networking applications: Edge computing, networking routers & gateways, network protocol converters.
4. Automotive Applications: Vehicle diagnostics, In-vehicle applications
5. Health Care and Medical Devices: Medical monitoring systems, patient data acquisition

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6. Embedded systems and consumer electronics
7. Data logging and measurement systems
8. Control systems
9. Security and surveillance systems.

7. Draw the block diagram of TMHC and explain each block.

Refer to 5th question.

8. Explain the basics of addressing modes of embedded system.

Addressing mode is the way of addressing data or operand in the instruction.

Some fundamental addressing modes used by most of the processors, are register addressing, direct addressing, immediate addressing & register indirect addressing

Addressing Modes in ARM (Embedded System):-

1. Register Addressing mode:

operand resides in a register specified in the instruction.

Ex:- MOV R_1, R_2 (Move content of R_2 to R_1)

SUB R_0, R_1, R_2 (Subtract R_2 from R_1 and store the result in R_0).

2. Relative Addressing mode:

Address is directly specified in the instruction.

Ex:- BEQ LOOP (Branch to LOOP if the zero flag is set).

3. Immediate Addressing mode:

Operand is directly embedded in the instruction

Ex:- SUB R0, R0, #1 (Subtract 1 from R0)

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MOV R0, #0xFF00 (store 0xFF00 in R0)

4. Register Indirect Addressing Mode:

Operand resides in memory, and the address is stored in a register.

Ex:- LDR R_1, [R_2] (Load R_1 with data from the address in R_2)

ADD R_0, R_1, [R_2] (Add R_1 to data pointed to by R_2 and store in R_0)

5. Register offset Addressing Mode:

Operand 2 is in a register, and an offset is applied.

Ex:- MOV R_0, R_2, LSL #3
(Shift R_2 left by 3 and store in R_0).

AND R_0, R_1, R_2, LSR R_3
(Logically AND R_1, with R_2 shifted right by R_3, store result in R_0)

6. Register-Based with offset Addressing Mode:

Effective memory address is calculated from a base address and an offset.

Offset can be an immediate value, register value or scaled register value

Sub-modes:

1. Pre-indexed Addressing:

Address calculated and used without modifying the base register

Ex:- LDR R_2, [R_3, #08] (load data from address R_3+08 into R_2)

STR R_1, [R_0, R_2] (store R_1 data to address R_0-R_2)

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2. Pre-indexed with write Back (Auto-indexing):
Calculated address is saved back into the base register.

Ex:- LDR R0, [R1, #4]! (load data from address R1+4 to R0, update R1 to R1+4)

STR R3, [R1, R2 LSL #4]! (store R3 to address R1 + (R2 << 4), update R1).

9. Explain the basics of instruction set of embedded system.

Basics of instruction set of embedded systems (ARM)

ARM instructions are categorized as follows:-

1. Data Movement Instructions:
Move data between registers or between memory and registers.

Ex:- MOV R1, R2 (Copy R2 Content to R1).

2. Data Processing Instructions:
Include arithmetic, logical, Comparision and multiply operations.

- Arithmetic :-
Instructions in arithmetic includes

ADD - Addition. ADD R0, R1, R2
SUB - Subtraction SUB R0, R1, R2
ADC - Add with Carry ADC R0, R1, R2, 1
SBC - Subtract with Carry, SBC R0, R1, R2, 1

- Logical instructions
AND - add operator
ORR - logical OR
EOR - Exclusive OR Ex:- AND R0, R1, R2
BIC - Bit clear.

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- Comparison:

CMP - Compare.

TST - Test

Ex:- CMP R1, R2

- Multiply:

MUL - Multiply

MLA - Multiply and accumulate

MUL R3, R1, R2

- Barrel shifter: Preprocesses operands before ALU operations

Eg: ADD R0, R1, R2, LSL #3

3. Branch Instructions:

Control program flow. Ex:- B LOOP (Brach to LOOP)

4. Load and Store Instructions:

- Transfer data between memory and registers.
- Load-Store Register: LDR R1, [R2] (load data from memory pointed by R2 to R1).
- Load-Store Multiple Registers:

LDM - Load Multiple

STM - Store Multiple

- Stack Instructions: push / pop data from stack.
- Swap Register - Memory: SWP (Swap memory content with a register).

5. Program Status Register (PSR) instructions:

Modify or read flags, interrupt enable bits or processor modes.

Ex:- Set condition flags (e.g. S suffix in ADDS).

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6. Exception generating Instructions
generate software interrupts or break points.

Ex: SWI (software interrupt).

Instruction syntax:

- format: Operation <Rd, Rn, Rm>

Rd - destination register
Rn - first source register
Rm - second source register

Ex: ADD R0, R1, R2

- Flags in CPSR (Current program status Register):

N - Negative, Z - zero, C - carry, V - overflow.

Immediate operands:

Allows 12-bit immediate values (range 0 - 4096)

Ex: MOV R0, #4096

Multiply Instructions:

MUL - Basic multiplication (32 bit result)

MLA - Multiply - Accumulate

Long multiplication for 64-bit:

UMULL - unsigned Multiply long
UMLAL - unsigned Multiply accumulate long
SMULL - Signed Multiply long
SMLAL - Signed Multiply Accumulate long

Barrel shifter:

Shifts or rotates operands before ALU execution

eg: ADD R3, R2, R1, LSL #3

LSL - logical shift left

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10. Write short notes on TM4C123X family and its targeted applications

TM4C123X	Temperature	85°C	105°C

ARM Cortex-M4 up to 80MHz	
FPU	MPU
NIVC	ETM	SWD/T

Memory
up to 256 KB Flash
up to 32 KB SRAM
2 KB EEPROM
ROM
DMA (32 Ch)

Power & Clocking
Precision Oscillator
RTC Battery-Backed Hibernate

System Modules
6x 32-bit Timer/PWM/CCP
6x 64-bit Timer/PWM/CCP
Systick Timer
2x Watchdog Timer

Debug
Real-time JTAG

Control Peripherals
2x Quadrature Encoder inputs
16x PWM Outputs

Comms Peripherals
8x UART
4x SSI/SPI
6x I²C
2x CAN
USB Full Speed (Host/Device/OTG)

Analog
2x 12 Ch, 12-bit ADCs 1MSPS
LDO voltage Regulator
3x Analog Comparators
Temp Sensor.

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[Matter same as 5th question]

11. Explain on-chip peripherals in embedded system.

In embedded systems, on-chip peripherals include components integrated directly into the microcontroller, which are broadly categorized into system, serial, analog and motion control peripherals.

System peripherals :-

1. Watchdog Timer: Ensures system reliability by resetting the CPU if a software malfunction causes a loop. It monitors if the program counter runs into infinite cycles, thereby safeguarding against system freeze.

2. Hibernation Module: Manages power consumption by powering down the microcontroller and peripherals when idle. It includes a Real-time Clock (RTC) for timed wake events and a battery-backed SRAM for saving processor state.

3. Direct Memory Access (DMA): Enables high-speed data transfers between memory and I/O devices without involving the CPU, ideal for handling large data transfers.

4. General-Purpose Input/Output (GPIO): Provides flexibility with up to 43 programmable I/O pins, supporting edge/level-sensitive interrupts, fast toggling, and usage in ADC sampling or µDMA transfers.

5. Timers: The General-purpose timer module (GPTM) enables time measurement, counting events and can trigger ADC conversions or µDMA transfers.

Serial peripherals :-

1. Universal Asynchronous Receiver-Transmitter (UART): Supports asynchronous serial communication for data,

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exchange.

2. Inter-Integrated Circuit (I2C): A synchronous, multi-master, multi-slave serial communication bus for connecting peripherals.

3. Controller Area Network (CAN): Facilitates real-time data exchange in automotive and industrial applications.

4. USB On-the-Go (OTG): Enables USB communications in host/device mode.

5. Synchronous Serial Interface (SPI/SSI): Supports high-speed full-duplex communication for peripherals.

Analog peripherals :-

1. Analog Comparator: Compares two analog voltages, generating a logical output or triggering an interrupt / ADC event

2. Analog-to-Digital Converter (ADC): Converts analog signal to digital values for processing. It samples the signal at high frequencies to minimize loss, encoding values into binary.

Motion Control peripherals :-

1. Pulse Width Modulation (PWM): Controls analog devices via digital signals, useful in applications like motor speed control.

2. Quadrature Encoder Interface (QEI): Tracks the position, direction and speed of rotating components, useful in motor control applications.

These peripherals collectively enhance embedded systems by enabling efficient communication, power management, analog processing & motion control essential for modern applications.

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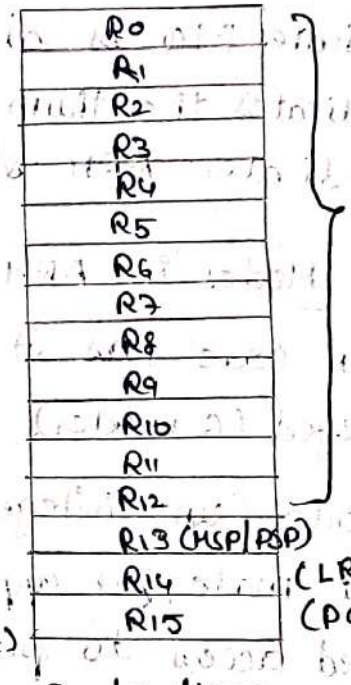
12. Explain Register set in Embedded System.

Register set :- Registers are for temporary data storage within processor architecture

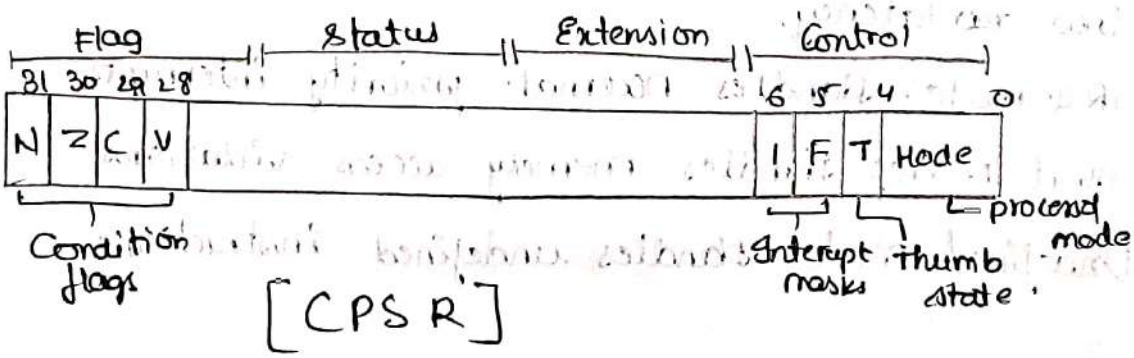
Register set in ARM processor

The ARM processor has 16 general-purpose registers (R0 - R15) and a 32 bit current program status register (CPSR) for the user mode.



- General Registers (R0 - R12):
  used for general-purpose operations.
- Special Registers:
- R13 (stack pointer):
  Holds the address of the top of the stack in the current processor mode
- R14 (Link Register):
  Saves the program counter (PC) content during executions / branch instructions.
- R15 (Program Counter):
  Points to the next instruction. In ARM state, it is aligned to word boundaries (least significant two bits are 0). For other states

  - Thumb state : Half-word (16bit) aligned
  - Jazelle state : Byte (8-bit) aligned.



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The CPSR is a 32-bit register that stores the processor's current state, including flags.

N - (Negative flag) = 1 indicates -ve result from ALU

Z - (Zero flag) = 1 indicates zero result from ALU

C - (carry flag) = 1, indicates ALU operation generated carry

V - (Overflow flag) = 1 indicates ALU operation overflowed

I = 1 ⇒ indicates IRQ is disabled 0 indicates IRQ enabled

F = 1 indicates FIQ is disabled 0 indicates FIQ enabled

T = 1 indicates the thum state is active

0 indicates ARM state is active

Operating Modes in ARM Core:

The ARM core has 7 operating modes, classified as privileged (6 modes) and unprivileged (1 mode).

1. User mode (unprivileged):

Default mode for application programs

Limited access to protected memory and CPSR control fields.

2. Privileged Modes:

- System mode: Runs OS kernel functions, shares user mode registers.
- Supervisor mode: Entered on reset or software interrupt used for OS services.
- FIQ mode: Handles high-priority, fast interrupts with low latency.
- IRQ mode: Handles normal-priority interrupts
- Abort mode: Handles memory access violations
- Undefined mode: Handles undefined instructions.

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Privileged modes allow full access to protected memory & CPSR control fields, while exceptions automatically switch the processor to a privileged mode.

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